

EC400 Probability and Statistical Inference

Pre-Arrival Quiz: Answers and Explanations

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Lecture 1: Foundations

1. Mark all answers which are true about populations, samples and statistical targets.

Correct answer(s): (a), (b), and (c).

- (a) A population contains all the units about which we want to learn; a sample contains the units we observe.
True. The population is the full group of interest, whereas the sample is the part for which data are observed.
- (b) The population defines the target, while the sample is used to estimate it.
True. The target is a population quantity; sample data provide the information used to construct an estimate of it.
- (c) A population target θ is fixed across repeated samples generated from the same population and design.
True. Repeated sampling changes the observed data and therefore the estimate, not the underlying target.
- (d) The same estimate must be obtained from every possible sample of a given size.
False. Different possible samples generally contain different observations and therefore produce different estimates.

2. Suppose $Y = DY(1) + (1 - D)Y(0)$, $P(D = 1) \in (0, 1)$ and $D \perp\!\!\!\perp (Y(1), Y(0))$. Mark all answers which are true.

Correct answer(s): (a), (b), and (d).

- (a) The average treatment effect is $ATE = E[Y(1) - Y(0)]$.
True. This is the definition of the population average treatment effect.
- (b) $E(Y \mid D = 1) - E(Y \mid D = 0) = ATE$.
True. Random assignment makes the treated and control groups comparable in their potential outcomes. Together with the observation rule for Y , the difference in observed conditional means equals the ATE.
- (c) For every individual in the experiment, both $Y(1)$ and $Y(0)$ are observed.
False. Only the potential outcome corresponding to the assigned treatment is observed for a given individual. The other potential outcome is counterfactual.
- (d) A difference in means can be causal under one design and merely descriptive under another.
True. The statistic is an estimation rule. Its causal meaning depends on the identification assumptions supplied by the design or model.

3. Consider repeated sampling from the same population under the same design. Mark all answers which are true.

Correct answer(s): (a), (b), and (c).

- (a) A sampling distribution describes an estimator across repeated samples or random assignments.
True. The sampling distribution records how the estimator would vary under repetitions of the data-generating or assignment process.
- (b) A standard error is the sampling standard deviation of an estimator, or an estimate of it.
True. The standard error quantifies repeated-sampling variability. In practice it is usually estimated from the observed data.
- (c) After the sample is observed, the realized estimate is a fixed number.
True. The estimator was random before sampling; its realized value is fixed once the data are observed.
- (d) An estimate is the rule $g(\cdot)$ rather than the number produced in the observed sample.
False. This reverses the terminology. The estimator is the rule; the estimate is its realized numerical value in the observed sample.

4. Let $\hat{\theta}_n$ be an estimator of a scalar target θ . Mark all answers which are true.

Correct answer(s): (a) and (b).

- (a) $\hat{\theta}_n$ is unbiased if $E(\hat{\theta}_n) = \theta$.
True. Unbiasedness is a finite-sample statement about the estimator's expectation.
- (b) $\hat{\theta}_n$ is consistent if, for every $\varepsilon > 0$, $P(|\hat{\theta}_n - \theta| > \varepsilon) \rightarrow 0$ as $n \rightarrow \infty$.
True. This is the standard definition of convergence in probability to θ .
- (c) Every unbiased estimator is consistent.
False. Unbiasedness alone does not force the estimator's variance to shrink with n . An unbiased estimator can therefore fail to be consistent.
- (d) Consistency requires $\hat{\theta}_n = \theta$ in every finite sample.
False. Consistency is an asymptotic property. It permits sampling error in every finite sample.

5. Assume the relevant second moments exist. Mark all answers which are true.

Correct answer(s): (a), (b), and (d).

- (a) $MSE(\hat{\theta}) = Var(\hat{\theta}) + Bias(\hat{\theta})^2$.
True. This is the bias-variance decomposition of mean squared error.
- (b) If $\hat{\theta}$ is unbiased, then $MSE(\hat{\theta}) = Var(\hat{\theta})$.
True. For an unbiased estimator the squared-bias term is zero.
- (c) An estimator with lower variance necessarily has lower MSE, regardless of its bias.
False. MSE depends on both variance and squared bias. A low-variance estimator can still have high MSE if its bias is large.
- (d) A biased estimator can have lower MSE than an unbiased estimator.
True. A modest increase in bias can be more than offset by a sufficiently large reduction in variance.

Lecture 2: Statistics I

6. Let X be a continuous random variable with cdf F and density f . Mark all answers which are true.

Correct answer(s): (a), (c), and (d).

- (a) $F(x) = P(X \leq x)$.

True. This is the definition of the cumulative distribution function.

- (b) $f(x) = P(X = x)$.

False. For a continuous random variable, a single point has probability zero. A density is probability per unit of the horizontal scale, not a point probability.

- (c) For $a < b$, $P(a < X \leq b) = F(b) - F(a) = \int_a^b f(t) dt$.

True. For a continuous distribution, interval probabilities are cdf differences and density integrals.

- (d) A cdf is non-decreasing, with limits 0 and 1 at the left and right tails respectively.

True. These are basic defining properties of every cdf.

7. Let X have finite variance and let a and b be constants. Mark all answers which are true.

Correct answer(s): (a), (b), and (d).

- (a) $E(aX + b) = aE(X) + b$.

True. Expectation is linear.

- (b) $Var(aX + b) = a^2 Var(X)$.

True. Adding a constant does not affect spread, while multiplication by a multiplies squared deviations by a^2 .

- (c) $SD(aX + b) = a SD(X)$ for every real value of a .

False. The correct formula is $SD(aX + b) = |a|SD(X)$. A standard deviation cannot be negative.

- (d) $Var(X) = E(X^2) - [E(X)]^2$.

True. Expanding $E[(X - E(X))^2]$ gives the stated computational formula.

8. Let X and Y have joint density $f_{XY}(x, y)$, and suppose $f_Y(y) > 0$ at the value under consideration. Mark all answers which are true.

Correct answer(s): (a), (b), and (d).

- (a) $f_X(x) = \int f_{XY}(x, y) dy$.

True. Integrating the joint density over all possible values of Y gives the marginal density of X .

- (b) $f_{X|Y}(x | y) = \frac{f_{XY}(x, y)}{f_Y(y)}$.

True. A conditional density is a normalized slice of the joint density.

- (c) $f_{XY}(x, y) = f_X(x)f_Y(y)$ for every pair of jointly distributed random variables.

False. The factorization into marginals is a special property of independence, not a general identity.

- (d) If X and Y are independent, then $f_{X|Y}(x | y) = f_X(x)$ wherever the conditional density is defined.

True. Under independence, observing Y does not change the distribution of X .

9. Assume all expectations and variances below exist. Mark all answers which are true.

Correct answer(s): (a), (b), and (c).

- (a) $E(X) = E[E(X | Y)]$.

True. This is the law of iterated expectations.

- (b) $Var(X) = E[Var(X | Y)] + Var(E[X | Y])$.

True. This is the law of total variance: total spread equals average within-group spread plus between-group spread in conditional means.

- (c) If X and Y are independent, then $E(X | Y) = E(X)$ almost surely.

True. Independence leaves the conditional distribution, and hence the conditional mean, unchanged.

- (d) If $E(X | Y) = 0$ almost surely, then X and Y must be independent.

False. A zero conditional mean imposes mean independence in one direction, but it does not imply full independence of the joint distribution.

10. Consider the distinctions among random sampling, random assignment and iid observations. Mark all answers which are true.

Correct answer(s): (a), (b), and (c).

- (a) Random assignment can identify a causal treatment comparison within the study sample.

True. Random assignment breaks systematic links between treatment and potential outcomes, supporting internal validity.

- (b) Random sampling helps connect a study sample to a target population.

True. Random sampling supports representativeness and therefore external validity for the target population.

- (c) An iid sample consists of observations that are mutually independent and have the same marginal distribution.

True. The abbreviation “iid” stands for “independent and identically distributed.”

- (d) Random assignment and random sampling are two names for the same requirement.

False. Assignment concerns how treatment is allocated; sampling concerns how units enter the study. They solve different problems.

Lecture 3: Statistics II

11. Let $Z, Z_1, \dots, Z_\nu$ be independent $N(0, 1)$ random variables and let $U \sim \chi_\nu^2$ be independent of Z . Mark all answers which are true.

Correct answer(s): (a), (b), and (c).

- (a) $Z^2 \sim \chi_1^2$.

True. A chi-squared variable with one degree of freedom is the square of one standard normal.

- (b) $\sum_{j=1}^\nu Z_j^2 \sim \chi_\nu^2$.

True. Degrees of freedom add when independent squared standard normals are summed.

- (c) $\frac{Z}{\sqrt{U/\nu}} \sim t_\nu$.

True. This ratio is the standard construction of a Student t random variable.

- (d) If independent $U_j \sim \chi_{\nu_j}^2$, then $U_1/U_2 \sim F_{\nu_1, \nu_2}$.

False. The chi-squared variables must first be divided by their degrees of freedom. The correct ratio is

$$\frac{U_1/\nu_1}{U_2/\nu_2} \sim F_{\nu_1, \nu_2}.$$

12. Mark all answers which are true about the existence of moments.

Correct answer(s): (a) and (d).

- (a) If $E(|X|^4) < \infty$, then $E(|X|^2) < \infty$.

True. A finite higher absolute moment implies all lower absolute moments are finite; this follows, for example, from Jensen's inequality.

- (b) If $E(|X|) < \infty$, then $E(X^2) < \infty$.

False. A distribution can have a finite first absolute moment but an infinite second moment.

- (c) Every probability distribution has finite moments of every positive order.

False. Heavy-tailed distributions can lack some or all higher moments.

- (d) If a right-tail density behaves like $f(x) \asymp c/x^2$ for large x , then its second moment is infinite.

True. In that tail, $x^2 f(x)$ stays of constant order rather than decaying, so its integral over the tail diverges.

13. Assume the expectations below exist. Mark all answers which are true.

Correct answer(s): (a), (b), and (c).

- (a) If g is convex, then $g(E(X)) \leq E[g(X)]$.

True. This is Jensen's inequality for convex functions.

- (b) If g is concave, then $g(E(X)) \geq E[g(X)]$.

True. The inequality reverses for concave functions.

- (c) Applying Jensen's inequality to $g(x) = x^2$ gives $E(X^2) \geq [E(X)]^2$.

True. Since x^2 is convex, Jensen yields this inequality, which is equivalent to the non-negativity of variance.

- (d) Equality in Jensen's inequality holds for every convex function and every random variable.

False. Equality holds in special cases, such as an affine function or a degenerate random variable, but not in general.

14. Consider the regression notation $y = X\beta + \varepsilon$ with n observations and k regressors, including an intercept if one is used. Mark all answers which are true.

Correct answer(s): (a), (b), and (c).

- (a) y and ε are $n \times 1$ vectors.
True. There is one outcome and one disturbance for each of the n observations.
- (b) X is $n \times k$ and β is $k \times 1$.
True. The matrix product $X\beta$ is therefore $n \times 1$, matching the dimensions of y .
- (c) Row i of X is x'_i , so the scalar equation is $y_i = x'_i\beta + \varepsilon_i$.
True. Stacking these n scalar equations produces the matrix equation.
- (d) $X'X$ is an $n \times n$ matrix.
False. $X'X$ is $k \times k$. In fact, $X'X = \sum_{i=1}^n x_i x'_i$.

15. Let $u = (u_1, \dots, u_n)'$ and let $\Omega = \text{Var}(u)$. Mark all answers which are true.

Correct answer(s): (a), (b), and (c).

- (a) Ω is symmetric and its diagonal entries are the variances $\text{Var}(u_i)$.
True. Covariance matrices are symmetric, and covariance of a variable with itself is its variance.
- (b) The off-diagonal entry Ω_{ij} is $\text{Cov}(u_i, u_j)$.
True. The off-diagonal entries record pairwise linear co-movement.
- (c) Homoskedastic and mutually uncorrelated errors have $\Omega = \sigma^2 I_n$.
True. Homoskedasticity gives the common diagonal value σ^2 ; lack of correlation sets all off-diagonal entries to zero.
- (d) If every off-diagonal covariance is zero, the components of u must be mutually independent.
False. Zero covariance does not in general imply independence, although it does under some special distributional assumptions such as joint normality.

Lecture 4: Hypothesis Testing

16. Consider a hypothesis test of $H_0 : \beta = \beta_0$. Mark all answers which are true.

Correct answer(s): (a), (b), and (d).

- (a) The null hypothesis is the default claim whose compatibility with the data is assessed.
True. Classical testing begins by treating H_0 as the maintained default and asks whether the evidence is sufficiently incompatible with it.
- (b) A p -value is the probability, calculated under H_0 , of observing a test statistic at least as extreme as the realized one.
True. This is the frequentist definition of a p -value.
- (c) A p -value is the posterior probability $P(H_0 \text{ is true} \mid \text{data})$.
False. A classical p -value conditions on the null being true; it does not assign a probability to the truth of the null.
- (d) The significance level α should be chosen before looking at the estimate.
True. Choosing α after seeing the result would allow the evidence to influence the decision rule and distort its advertised error rate.

17. For a standard two-sided level- α test and its corresponding $(1 - \alpha)100\%$ confidence interval, mark all answers which are true.

Correct answer(s): (a), (b), and (d).

- (a) The test rejects when the p -value is smaller than α .
True. The p -value reports the smallest conventional significance level at which the observed evidence would lead to rejection.
- (b) The test rejects $H_0 : \beta = \beta_0$ when β_0 lies outside the corresponding confidence interval.
True. A confidence interval is the set of null values not rejected by the corresponding family of two-sided tests.
- (c) Failure to reject H_0 proves that H_0 is true.
False. Failure to reject means the evidence was not sufficiently extreme under the decision rule. It is not a proof of the null.
- (d) The critical value depends on the reference distribution and the chosen significance level.
True. Different reference distributions and different tail probabilities produce different cut-offs.

18. Suppose $\hat{\beta} = 0.12$, $\widehat{se}(\hat{\beta}) = 0.05$, and we test $H_0 : \beta = 0$ using a large-sample standard normal reference distribution. Mark all answers which are true.

Correct answer(s): (a), (b), and (c).

- (a) The test statistic is $t = (0.12 - 0)/0.05 = 2.40$.
True. The test statistic is the distance from the null value measured in estimated standard errors.
- (b) At the 5% two-sided level, the test rejects because $|2.40| > 1.96$.
True. The observed statistic lies beyond the two-sided 5% standard-normal critical values ± 1.96 .
- (c) The approximate 95% confidence interval is $[0.022, 0.218]$, up to rounding, and it excludes zero.
True. $0.12 \pm 1.96(0.05) = 0.12 \pm 0.098$, which gives the stated interval.
- (d) The two-sided p -value is greater than 0.05.
False. Since the statistic exceeds the 5% critical value in absolute value, the corresponding two-sided p -value is below 0.05.

19. Mark all answers which are true about testing errors and power.

Correct answer(s): (a), (b), and (c).

- (a) A Type I error is rejection of a true null hypothesis.

True. The significance level controls the probability of this error under the null.

- (b) A Type II error is failure to reject a false null hypothesis.

True. This is the error of missing a real departure from the null.

- (c) Power at a particular alternative is one minus the probability of a Type II error at that alternative.

True. Power is the probability of correctly rejecting the null when the stated alternative is true.

- (d) Lowering α while holding the design, sample size and alternative fixed generally increases power.

False. A lower α moves the rejection threshold farther into the tail, making rejection harder and generally reducing power.

20. Compare one-sided and two-sided tests at the same significance level α . Mark all answers which are true.

Correct answer(s): (a), (b), and (d).

- (a) A one-sided test places the full rejection probability α in one tail.

True. Unlike a two-sided test, it does not divide α across two tails.

- (b) A one-sided test can have greater power against alternatives in its pre-specified direction.

True. Concentrating the rejection region in the relevant tail lowers the cutoff in that direction and can increase power there.

- (c) It is valid to choose the direction of a one-sided test after observing the sign of the estimate.

False. The direction must be justified and chosen before seeing the evidence; otherwise the nominal error rate is not the one advertised.

- (d) A two-sided alternative such as $H_1 : \beta \neq \beta_0$ treats sufficiently large departures in either direction as evidence against H_0 .

True. A two-sided test rejects for estimates far above or far below the null value.

Lecture 5: Asymptotics

21. Let $\hat{\theta}_n$ be an estimator and let θ be a constant. Mark all answers which are true.

Correct answer(s): (a), (b), and (c).

- (a) Consistency means $\hat{\theta}_n \xrightarrow{p} \theta$.

True. This is the definition of consistency.

- (b) Convergence in probability implies convergence in distribution.

True. Convergence in probability is stronger than convergence in distribution.

- (c) Convergence in distribution to the constant θ implies convergence in probability to θ .

True. When the distributional limit is a constant, convergence in distribution and convergence in probability coincide.

- (d) Convergence in distribution to a non-degenerate random variable is the same thing as consistency for θ .

False. Consistency requires concentration on the fixed target, not convergence to a random, non-degenerate limit.

22. Let $X_1, X_2, \dots$ be iid with $E(|X_i|) < \infty$ and $E(X_i) = \mu$. Mark all answers which are true.

Correct answer(s): (a), (c), and (d).

- (a) The law of large numbers gives $\bar{X}_n \xrightarrow{p} \mu$.

True. The sample average converges in probability to the population mean under the stated conditions.

- (b) The law of large numbers by itself says that $\sqrt{n}(\bar{X}_n - \mu)$ is approximately normal.

False. Approximate normality of the scaled error is the role of a central limit theorem, not the law of large numbers.

- (c) Dependent data can also satisfy a law of large numbers, but additional conditions are needed.

True. Time-series and panel observations need not be independent; suitable weak dependence and regularity conditions can still deliver an LLN.

- (d) Law-of-large-numbers arguments are a principal route to proving consistency of sample analogs.

True. Many estimators are averages or functions of averages, so LLNs turn sample quantities into their population counterparts.

23. Let $X_1, X_2, \dots$ be iid with $E(X_i) = \mu$ and $0 < \text{Var}(X_i) = \sigma^2 < \infty$. Mark all answers which are true.

Correct answer(s): (a), (c), and (d).

- (a) $\frac{\sqrt{n}(\bar{X}_n - \mu)}{\sigma} \xrightarrow{d} N(0, 1)$.

True. This is the classical iid central limit theorem.

- (b) $\bar{X}_n$ is exactly normally distributed for every n , whatever the distribution of X_i .

False. Exact normality holds, for example, when the observations themselves are normal. The CLT otherwise gives an approximation for large n .

- (c) The standard deviation of $\bar{X}_n$ is of order $n^{-1/2}$.

True. $\text{Var}(\bar{X}_n) = \sigma^2/n$, so its standard deviation is $\sigma/\sqrt{n}$.

- (d) The classical finite-variance central limit theorem need not apply when $\text{Var}(X_i) = \infty$.

True. Infinite-variance data can have different limiting behavior and require other normalizations or limit laws.

24. Suppose $\sqrt{n}(\hat{\theta}_n - \theta) \xrightarrow{d} N(0, V)$ with $0 < V < \infty$. Mark all answers which are true.

Correct answer(s): (a), (b), and (c).

(a) The stated asymptotic normality implies $\hat{\theta}_n \xrightarrow{p} \theta$.

True. The scaled estimation error remains stochastically bounded, so the unscaled error is of order $n^{-1/2}$ and converges to zero in probability.

(b) For large n , it suggests the approximation $\hat{\theta}_n \approx N(\theta, V/n)$.

True. Rearranging the asymptotic statement gives the usual large-sample normal approximation.

(c) If $\hat{V} \xrightarrow{p} V$, an approximate 95% confidence interval is $\hat{\theta}_n \pm 1.96\sqrt{\hat{V}/n}$.

True. Consistency of $\hat{V}$ and Slutsky's theorem produce the standard large-sample interval.

(d) Asymptotic normality guarantees that the estimand has the intended causal or economic interpretation.

False. Asymptotic theory describes estimation error around the estimand. Identification is a separate question about what that estimand means.

25. *Non-examinable extension.* Mark all answers which are true about continuous mapping and Slutsky's theorem.

Correct answer(s): (a), (b), and (d).

(a) If $X_n \xrightarrow{p} c$ and g is continuous at c , then $g(X_n) \xrightarrow{p} g(c)$.

True. This is the continuous mapping theorem for convergence in probability.

(b) If $X_n \xrightarrow{d} X$ and $Y_n \xrightarrow{p} c$, then $X_n + Y_n \xrightarrow{d} X + c$.

True. This is one of the standard forms of Slutsky's theorem.

(c) If $Y_n \xrightarrow{p} 0$, then X_n/Y_n always has a well-behaved limiting distribution.

False. Division is not continuous at a zero denominator. No general conclusion is available without additional structure.

(d) Slutsky's theorem often justifies replacing an unknown scale parameter by a consistent estimator of it.

True. A consistently estimated standard error can be substituted into an asymptotic pivot without changing its limiting distribution under the usual conditions.